UNIT-2

1. Examine the validity of the following Boolean functions.

(A). $(A+B+C)(A+B+\overline{C})=A+B$.

(b). $Z\overline{X}$+ZXY = ZX.

2. What do you mean by K-Map? Simplify the following

Expression by K-Map:

(A). $\boxed{\overline{A}}\,B\overline{C} + AB\overline{C} + A\overline{BC}$

(B). $A\overline{B}C+AB\overline{C}+\overline{A}BC+ABC+A\overline{BC}$

3. i. explain Minterm and Maxterm

ii. Differentiate between Boolean alzebra and ordinary alzebra.

iii.what do you mean by switching algebra?

Iv .what are Boolean postulates

4. what is venn diagram? Draw venn diagram for AND ,OR,NOT operations . also prove second absorption law $a+(\overline{a}.b)$=a+b using venn diagram

b.) simplify $\overline{X}\overline{Y}$+X+XY

c. express $F=A+\overline{B}.C$ is sum of minterms

d. explain exclusive OR and Exclusive NOR withtable and draw circuit diagram

e."AND OR,NOT gates are logically complete"discuss

5. a.) Convert the expression $F=(\overline{X}+Y)X+Z(Y+Z)$ into standards POS form

b.) examine the validity of:

$(XY).(YZ)=(\overline{X}+\overline{Y}).(\overline{Y}+\overline{Z})$

6. Define Boolean algebra and write postulates of Boolean alegebra.
7. State and prove De-Morgan's law
8. Explain two canonical forms of Boolean expression
9. What is truth table ,write truth table for expression X.(Y+Z)+XY
10. What NAND gate is used to perform function of AND,OR,NOT gate
11. Solve Boolean algebra

    i. $(X+Y)(XZ+Z)(\overline{Y}+XZ)=\overline{X}YZ$

    ii. $abc+\overline{a}bc+a\overline{b}c+ab\overline{c}$

    iii ab+bc+ca

iv (a+b)(ac+c)$(\overline{b + a\,c})$=$\overline{a}$bc

12. Draw and labl 4 variable K-Map and solve for four corners

b.solve using k- map

Z=$\sum 1,3,5,7$

Z= $\sum 0,1,10,11,15$ & $\sum 4,5,8,14$

Z=$\prod 0,2,4,6$

Z=$\sum 0,1,9,10,11$+$\sum_0 2,3,8,12$

Z=$\sum 1,2,3,7,10,11,12,15$

Z= $\sum 4,5,6,7,11,12,15$